



Endozoochory by the guild of ungulates in Europe's primeval forest



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ABSTRACT

Seed dispersal by large herbivores helps to maintain the plant species diversity. Forest management and game management influence species composition and the structure of ungulate communities. This may disrupt ecological processes depending on services provided by animals. In European secondary forests, sets of plant species dispersed by guilds of herbivores overlap, and one species, red deer *Cervus elaphus*, may replace all other seed vectors in this process, although the proportion of dispersed forest species is very low. We hypothesize that some forest plant species with present-day low dispersal abilities were in the past dispersed by large ungulates which have since become extinct. In our opinion, in natural forests, where guilds of herbivores were complete, seed vectors dispersed different sets of plant species. We studied endozoochoric seed dispersal by European bison *Bison bonasus*, Eurasian moose *Alces alces*, red deer *C. elaphus*, roe deer *Capreolus capreolus* and wild boar *Sus scrofa* in the close-to-natural ecosystem of Białowieża Forest, Poland. Each of the studied ungulates dispersed a higher number of plant species than reported in any previous papers. This can be attributed to the high biodiversity of the study site, and to different methodological approach: year round collection of samples and 3-years long time frame for germination. Red deer and European bison were the most efficient seed dispersers – dispersing the richest sets of plant species, including ancient forest species, and the highest seed numbers. Moose and wild boar were the least efficient. Sets of plant species dispersed by each vector overlapped only partly. Our study stresses the importance of the completeness of the guild of seed dispersers for plant dispersal in forest habitats. Over half of overall plant species emerged from dung samples were dispersed by European bison – a species exterminated in most European forests centuries ago. The high number of plant species dispersed endozoochorically by the studied guild of animals, combined with the low similarity between sets of dispersed plant species, and the high share of species dispersed exclusively by only one of the vectors allows the conclusion that: (1) the dispersal efficiency of ungulates differs, but all species play a unique and important role in plant dispersal; (2) the species diversity of the guild of ungulates is very important for plant dispersal in temperate forests; and (3) seed dispersal in European forests has been seriously impoverished since their guilds of herbivores became deficient.

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1. Introduction

Primeval forests are almost extinct in Europe – only 0.2% of European forests still carry some primordial features (Bengtsson et al., 2000). They are among the most threatened ecosystems worldwide (Foley et al., 2005), and even the best-preserved patches, like Białowieża Primeval Forest (BPF) on the Polish/Belarusian border have been changed considerably (Bobiec, 2012). The intensity of forest management, including game management, influences species composition and the structure of

ungulate communities. This in turn may disrupt ecological processes that depend on services provided by these animals (Foley et al., 2005). Seed dispersal is such a service, which allows plant migrations between patches of suitable habitats isolated by a hostile landscape matrix (e.g. Howe and Smallwood, 1982; Janzen, 1984; Pakeman, 2001). Endozoochoric plant dispersal maintains the species diversity of unspecialized vascular plants with seeds without morphological adaptations to long distance dispersal. In north-western and central Europe approximately 20% of the forest flora is characterized as 'ancient forest plant species', which have low dispersal and colonizing abilities (Hermy et al., 1999). Animal-mediated seed dispersal enables plants to escape from the negative effects of the mother plants' neighbourhood (e.g. increased pathogen and predation pressure), to colonize new habitats (Howe and Smallwood, 1982), and to reach adequate

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microhabitats for their development (Wenny, 2001; D'hondt et al., 2012).

Białowieża Primeval Forest (BPF) is one of the best preserved fragments of temperate forest on the European lowland (Faliński, 1986). The unique features of Białowieża's ungulate community are: (1) presence of all species of ungulates that are native to the Central-European lowland forests, and (2) lack of alien herbivore species which could disrupt the functioning of the native animal communities by competing for food resources and causing antagonistic interactions (Jędrzejewska and Jędrzejewski, 1998). BPF is inhabited by five ungulate species: wild boar (*Sus scrofa* L.), roe deer (*Capreolus capreolus* L.), red deer (*C. elaphus* L.), Eurasian moose (*Alces alces* L.) and European bison (*Bison bonasus* L.). All these species are known to play some role in plant dispersal (e.g. Heinken et al., 2002; Eycott et al., 2007; Jaroszewicz et al., 2009), although it is difficult to estimate their joint importance and the relative importance of each species without studies carried out for all of them in the same site.

The only comprehensive work considering the role of not a single species but a guild of mammalian herbivores in endozoochoric seed dispersal in temperate forest ecosystems was carried in Thetford Forest (eastern England; Eycott et al., 2007). The results suggest that the richness of the herbivore community does not influence its plant dispersal capacity. Sets of vascular plant species dispersed by red deer, fallow deer (*Dama dama* L.), roe deer, muntjac deer (*Muntiacus reevesi* Ogilby), hare (*Lepus europaeus* Pallas) and rabbit (*Oryctogalus cuniculus* L.) overlap with those dispersed by the red deer/fallow deer species complex, which dispersed 96 out of 101 plant species emerged from dung samples (Eycott et al., 2007). Another study, carried out in north-eastern Germany, was not concentrated on the community of plant dispersers but took into consideration a wide range of seed vectors: wild boar, roe deer, hare and marten (*Martes* sp.). The number of reported plant species dispersed endozoochorically by these four species was very low (38 plant species), and most of them were common field weeds (Heinken et al., 2002). Similar results were obtained by Mouissie (2004) for a guild of three deer species in the Netherlands. However, the most efficient disperser there was fallow deer, which, together with red deer and roe deer, dispersed 14 plant species.

These three studies show that seed vectors are not specialized but disperse similar sets of plant species (Heinken et al., 2002; Mouissie, 2004; Eycott et al., 2007) overlapping with the set dispersed by red deer/fallow deer. It should be noted that these studies were carried out in a mosaic landscape of small forest complexes surrounded by agricultural land and semi-natural open habitats. In effect all studied species utilized a similar mosaic of habitats. As a consequence sets of dispersed plant species were similar between vectors. In BPF, habitats occupied by different ungulate species overlap to a small extent (Faliński, 1986; Jędrzejewska and Jędrzejewski, 1998), and thus one may expect that the diets of animals and the species composition of dispersed seeds would overlap only to the extent resulting from their habitat preferences. Moose habitat preferences (wet forests and boggy river valleys) overlap only slightly with roe deer preferences (meso- and oligotrophic mixed forests, often at the edge of river valleys or clearings) and with wild boar, which seasonally frequent wet habitats. Bison and red deer show preference towards meso- and eutrophic deciduous forests. Wild boar inhabit very diverse types of forests, preferably patches with old-growth woodstands (Faliński, 1986; Jędrzejewska and Jędrzejewski, 1998).

Seed dispersal efficiency of herbivores is also influenced by their body size. Larger animals consume larger amounts of seeds (Malo and Suárez, 1995) and disperse them over larger distances because their home ranges are larger (Haskell et al., 2002). How-

ever, the time of food retention in the gut is longer for larger animals, which may cause lower seed viability in their faeces.

Therefore, in this paper we hypothesize that in the past, before herbivore guilds in Europe were depleted by overhunting, and before forest ecosystems were fragmented and unified by forest management, sets of plant species dispersed by large ungulates were much richer and did not overlap to so high a degree as reported by Heinken et al. (2002), Mouissie (2004), Eycott et al. (2007). Thus, in natural forest each herbivore species would play an important role in the process of plant dispersal, and the species composition of animal guilds would influence rates of seed migrations and viability of plant populations. In this paper we address particularly the following questions: (1) How important is each of the ungulate species for endozoochorous plant dispersal? i.e. how many plant species and how many seeds do they disperse? (2) To what extent do sets of plant species dispersed by seed vectors in close-to-natural forest overlap? (3) Is the diversity of the guild of ungulates important for plant dispersal? (4) Can ungulate species replace each other in the process of endozoochoric seed dispersal?

2. Materials and methods

2.1. Study area

We conducted our study in Białowieża Primeval Forest (BPF), a large forest massif straddling the border between Poland and Belarus (52°42'–52°59'N; 23°43'–23°56'E). The entire forest covers an area close to 1500 km², although the border fence has isolated populations of large mammals for the last 30 years. The Polish (western) part of the Forest covers approximately 625 km², with close to 225 km² within the limits of the national park and nature reserves. It is covered by mixed deciduous forests (over 60% of the area) and with only a 5% proportion of open areas (Sokołowski, 2004). The forest ecosystem in the core area (ca. 47 km²) of the national park is the most pristine piece of forest in Central Europe (Faliński, 1986).

The guild of ungulates in BPF comprises five species, the most common of which are red deer and wild boar (respectively, 3228 and 2330 individuals in 2011; Forest Research Institute unpublished data). Moose is the rarest ungulate of the forest (approximately 60–80 individuals; authors' own unpublished estimation). The Roe deer population was estimated at approximately 920 individuals (Forest Research Institute unpublished data) and European bison at 481 individuals (Białowieża National Park unpublished data).

There are over 1100 vascular plant species recorded in BPF. Of these approximately 670 species constitute native components of natural forest plant communities while the remaining species colonized the forest as a result of human-caused disturbances (Sokołowski, 1995, updated).

2.2. Dung collection and treatment

Dung sampling was year-round and evenly distributed over time. Samples of faeces of bison and all deer species were collected fortnightly (one collective sample on each date). Samples of wild boar droppings were collected approximately fortnightly but only each second month (i.e. at the beginning, in the middle and at the end of January, March, May, July, September and November), five samples on each date (Table 1). We consider that the differences in the dates of sampling did not influence the results considerably as the time of fructification and the period of fruit retention varies among and between individual plants, as well as between plants growing in different sites and habitats.

Table 1

Schemes of dung sampling in Białowieża Primeval Forest (northeastern Poland).

Ungulate species	Sampling frequency	Start and end of sampling	No. of samples	Mean pellet/pat volume [l] ± SD	Mean sample volume [l] ± SD
European bison <i>Bison bonasus</i>	Fortnightly	Sept 2005 Sept 2006	46	1.57 ± 0.72	2.5 ± 0.00
Eurasian moose <i>Alces alces</i>	Fortnightly	Jan 2008 Jan 2009	35	0.443 ± 0.130	0.8 ± 0.098
Red deer <i>Cervus elaphus</i>	Fortnightly	Sept 2007 Sept 2008	80	0.112 ± 0.042	0.54 ± 0.197
Roe deer <i>Capreolus capreolus</i>	Fortnightly	Sept 2007 Sept 2008	33	0.033 ± 0.012	0.34 ± 0.220
Wild boar <i>Sus scrofa</i>	Fortnightly every second month	Jan 2009 Dec 2009	90	0.25 ± 0.05 ^a	0.25 ± 0.05

^a Only large pellets of wild boar were selected for collection.

Samples were collected along transects in all types of forest habitat visited by the studied herbivores. To extend the range of plant species potentially dispersed by the studied ungulates, not only forests but also clearings in the forest were searched for dung samples. Unevenness in the size of the samples and sample sets (Table 1) results from the density of ungulates (influencing the chances that dung will be found) and the diversity of the ecosystems where their faeces were present.

Only fresh dung (wet and intact) was collected in order to avoid contamination by seed rain. Faeces from different individuals were consolidated to fill the sampling container. Bison faeces were pooled together to obtain a 2.5 l sample. The lowermost layer of the bison dung was left behind to avoid contamination by seeds from the soil surface. All deer species' pellet groups were collected to fill tightly a 1.0 l container. However, only moose and red deer samples were close to even in volume (0.8 l and 0.4 l, respectively), while roe deer sample volume varied according to the number of pellet groups found. Wild boar samples were collected to fill a 0.5 l container and were also close to even in volume (0.3 l; Table 1). To calculate the real volume of each sample we measured the volume of water poured into the container with the sample and deducted it from the container's volume.

In the laboratory, faeces were mixed with sterilized sand (kept at 110 °C for 48 h) in a ratio of 1:3 (1:2 for bison dung) and spread onto trays in an approximately 4–6 cm-thick layer. The sample trays and control trays (with pure sterilized sand) were kept in a greenhouse and watered accordingly. They were left for wintering in the greenhouse to create conditions similar to natural stratification. Samples were not watered during winter in order to avoid premature germination. Each tray's substratum was stirred every time seedlings were removed during observations.

2.3. Estimation of viable seed content

Emerging seedlings were counted during three consecutive vegetation seasons following dung collection (Table 1) and removed from trays as soon as their determination was possible. Individual seedlings that could not be identified during two–three consecutive observations were replanted in pots and grown in a greenhouse or in the experimental garden until their identification could be verified. *Viola riviniana* (Rchb.) and *Viola reichenbachiana* (Jord. ex Boreau), and also *Carex spicata* (Huds.) and *Carex muricata* (L.) could not be identified accurately and were treated as collective taxon *V. riviniana/reichenbachiana* and *C. spicata/muricata*. For simplicity, we refer to all taxa as species, including those seedlings that could only be identified to the genus (e.g. *Agrostis* spp., *Alchemilla* spp., *Betula* spp., *Carex* spp., *Epilobium* spp., *Juncus* spp.). Undetermined dicotyledonous and graminoid seedlings that died before an accurate determination was possible were excluded from analyses, and they constituted 1.7% of seedlings overall. Seed-

lings of only two plant species (*Betula* sp. and *Salix* sp.) emerged on pure sand in control trays. In our opinion seeds of *Betula* and *Salix* can pass undamaged through the digestive tracks of ungulates. That's the reason to consider them as "of endozoochoric origin" during the first year of germination (even if some seedlings of these genera emerged on purified sand in control trays, suggesting contamination by airborne seeds). In subsequent years all seedlings of these taxa were classified as contamination, and excluded from further analyses, because their seeds are transient, i.e. their viability is limited to one year. It is very unlikely that any seedling emerged during the second and third years from trays containing samples originating from dung. Plant nomenclature follows Mirek et al. (2002).

2.4. Data analysis

Species with viable seeds in faeces were categorized by habitat groups (forest, grasslands, forest/grassland and synanthropic) according to their occurrence in plant communities (Matuszkiewicz, 2001) and distribution in BPF documented by Sokołowski (1995). Some species that occurred with similar frequency in grassland and forest communities were assigned to the forest/grassland habitat group. Segetal, ruderal, invasive and cultivated species were considered synanthropic. Differences among the studied ungulates in the share of plant species originating from different habitat groups were determined using chi-square analysis (low number of groups). Plant dispersal strategies were classified using a variety of sources (e.g. Hill et al., 2004; Bojňanský and Fargašová, 2007; Grime et al., 2007).

The species richness of plants emerging from dung samples of different ungulates was compared by sample-based rarefaction. The plant species' compositions between seed vectors were compared using the ANOSIM procedure (ANalysis Of SIMilarity), which is a non-parametric test of significant difference between two or more groups, based on a distance measure (Clarke, 1993). It calculates a statistic *R*, ranging from −1 to +1. A large positive *R* signifies dissimilarity between groups (Hammer et al., 2001). To estimate the true species richness of dispersed plant sets we used Chao2 estimators of total species richness. They usually perform better than other estimators (Colwell and Coddington, 1994).

Differences in the mean number of species per 100 ml of faeces and differences in mean seedling abundance per 100 ml of faeces between ungulate species were tested by the Kruskal–Wallis test (data distribution non-normal). The frequency of plant species in dung samples was calculated as the proportion of samples with viable seeds. The frequency of plant species in forest is the proportion of forest districts in which species were recorded by Sokołowski (1995). Frequencies of plant species in samples and in the forest had non-normal distribution and were compared by use of the non-parametric Wilcoxon signed-rank test with the

Monte Carlo estimate for P (100,000 random permutations). The number of analysed species was limited to 167 (out of 191 which emerged from samples) because Sokołowski (1995) did not record several alien and synanthropic species.

We executed regression analysis to check the influence of seed traits (seed length, width and mass) on relative dung seed density in herbivore dung. Seed traits data were obtained from the Kew seed information database (Kew, 2011) and from Bojňanský and Fargašová (2007). All tests were performed using PAST 1.58 software (Hammer et al., 2001) with the exception of the calculation of Chao2 estimators and rarefaction analysis, performed using Estimates 8.2.0 (Colwell, 2006).

3. Results

3.1. Seed dispersal

A total of 14,154 seedlings of 191 species emerged from 284 dung samples of the studied five ungulate species (Table 2; Appendix A). Sixty-five percent of all seedlings recorded were of one species: *Urtica dioica* L. (9293 seeds). A large share of species (31%) occurred only in a single dung sample, and 71% of species produced 10 or fewer seedlings, while 11.5% of dung samples did not produce any seedlings.

The studied vectors dispersed seeds of plants from 35 families. The highest number of species belonged to Poaceae, Cyperaceae and Asteraceae (36, 18 and 17 species, respectively), while 34% of families were represented by one species only (Table 3). The number of plant species dispersed by the studied guild of ungulates was correlated with the species richness of their families in BPF (Spearman's $r_s = 0.751$; $P < 0.0001$). However, some families rich in species in the Forest were not recorded from any sample (e.g. Orchidaceae), or were represented only by single species (Apiaceae, Boraginaceae, and Liliaceae; Table 3).

The frequency of plant species in all dung samples of all studied ungulates differed from their frequency in BPF (the Wilcoxon test; $z = 10.42$, $P < 0.0001$). Similarly, the frequency of plant species in the dung of each studied ungulate species differed from their frequency in BPF ($P < 0.05$; z factor ranging between 9.07 for bison and 11.02 for moose and wild boar). Only *Deschampsia caespitosa* (L.) P.B., *Rubus idaeus* L., *U. dioica* and *Veronica chamaedrys* L. were common between the 10 most frequent species in the samples and the ten most frequent species in BPF (only species recorded from dung samples were considered).

Close to half of all species emerged from faecal samples were those with seeds morphologically not adapted to any type of long distance dispersal, and over one quarter of species were adapted for epizoochory. The share of each of these two groups was significantly higher in comparison to the anemochores and other dispersal groups (Mann–Whitney test $P < 0.04$). Differences between endozoochores and epizoochores were not significant. The vector species did not differ by the share of dispersed plant species

grouped by dispersal adaptations (all Mann–Whitney pairwise comparisons $0.9 > P > 0.06$; Fig. 1).

Grassland plants (37%) and forest plants (27%) had the highest proportion between the species recorded on the studied dung samples. Notably high (17% in average) was the share of species of synanthropic habitats, ranging between 23% for moose and 12% for bison. The wetland species had the lowest share (only 2%). The ungulate species did not differ by share of the habitat of origin of the dispersed plant species (Chi² tests for all pairs $P > 0.2$; Fig. 2).

3.2. Differences between vectors

Over 95% of red deer, bison and roe deer dung samples contained seeds, while only 60% of wild boar and 51% of moose faecal samples did. Bison dispersed the highest number of viable seeds and species per defecation unit (pat). Faeces of red deer and roe deer included the highest mean number of viable seeds per 100 ml of dung (20.5 ± 29.3 (SD) and 15.1 ± 23.2 (SD), respectively) and plant species per 100 ml of dung (1.6 ± 1.8 (SD) and 2.0 ± 2.4 (SD), respectively). The lowest means were recorded for moose dung (0.4 ± 0.8 (SD) seeds and 0.1 ± 0.2 (SD) species per 100 ml of dung) and wild boar dung (1.2 ± 3.1 (SD) seeds and 0.05 ± 0.05 (SD) species per 100 ml of dung), with noticeable low means also for bison faeces (3.7 ± 5.6 (SD) seeds and 0.4 ± 0.3 (SD) species per 100 ml of dung). Within-group variability of mean number of seeds per 100 ml of faeces and mean number of species per 100 ml of faeces was high (Kruskal–Wallis test $H = 124.4$; $P < 0.0001$ and $H = 184.5$; $P = 0.0001$, respectively). All studied ungulate species differed in mean number of viable seeds per 100 ml of dung (Mann–Whitney pairwise comparisons $P < 0.05$) except for red deer compared to roe deer ($P = 0.22$). Similarly, ungulates differed in mean number of dispersed species per 100 ml of dung (Mann–Whitney pairwise comparisons $P < 0.0001$) except for two pairs: red deer – roe deer ($P = 0.72$) and moose – wild boar ($P = 0.19$).

The composition of dispersed plant species varied among all vectors (ANOSIM global: $P < 0.0001$) and between each pair of vectors (ANOSIM for all pairwise comparisons $P < 0.002$) except for *A. alces* compared to *S. scrofa* (ANOSIM $P = 0.608$; $R = 0.04$; Table 4). The species composition of plants dispersed by moose was the least similar to any other species ($0.24 < R < 0.65$ for all pairwise comparisons), with the greatest dissimilarities noted for moose compared to red deer, and moose compared to bison ($R = 0.65$ and $R = 0.38$, respectively; Table 4).

Red deer and European bison dispersed the highest number of species: 71.7% and 57.1% of all species dispersed by the studied guild of ungulates, respectively. The lowest number of plant species was dispersed by moose and wild boar (14.3% and 12.6%, respectively). Cumulative species richness curves (Fig. 3) showed that red deer and bison are the most efficient seed dispersers, while moose and wild boar are the least efficient between the studied ungulates. Chao2 estimators of total species richness showed

Table 2
Dispersal characteristics of the guild of ungulates in Białowieża Primeval Forest (northeastern Poland), comprising: European bison *Bison bonasus*, Eurasian moose *Alces alces*, red deer *Cervus elaphus*, roe deer *Capreolus capreolus* and wild boar *Sus scrofa*; ancient forest species list follows Dzwonko and Loster (2001).

	European bison	Eurasian moose	Red deer	Roe deer	Wild boar	Total
No of dispersed species	109	27	137	64	24	191
Chao2 estimator of the total species richness (95% confidence intervals in the brackets)	132 (119–161)	51 (35–100)	198 (168–256)	94 (77–134)	47 (30–105)	N/A
No of exclusive species	32	5	41	8	2	88
No of exclusive species recorded in more than 1 sample	15	0	11	2	0	28
No of dispersed ancient forest species	20	1	15	8	3	25
No of dispersed families	21	16	30	22	15	35

Table 3

Number of plant species, grouped by plant families, dispersed by the guild of ungulates in Białowieża Primeval Forest (northeastern Poland); families sorted according to total species richness in dung samples; species number in Białowieża Primeval Forest according to Sokółowski (1995).

Plant family	European bison <i>Bison bonasus</i>	Eurasian moose <i>Alces alces</i>	Red deer <i>Cervus elaphus</i>	Roe deer <i>Capreolus capreolus</i>	Wild boar <i>Sus scrofa</i>	Total	Białowieża Forest
Poaceae	22	7	29	14	4	36	85
Cyperaceae	15	2	9	3	3	18	66
Asteraceae	2	1	15	6	0	17	108
Fabaceae	9	0	8	5	0	14	54
Onagraceae	8	0	8	4	0	11	15
Polygonaceae	9	1	8	2	2	11	24
Caryophyllaceae	8	2	9	5	0	10	43
Lamiaceae	4	1	6	3	1	8	41
Juncaceae	6	2	3	3	2	7	18
Scrophulariaceae	6	1	4	2	1	7	41
Rosaceae	4	3	5	2	1	6	59
Rubiaceae	2	0	4	1	3	5	16
Brassicaceae	0	0	3	2	1	4	40
Ranunculaceae	3	0	3	0	1	4	35
Betulaceae	1	1	3	1	1	3	8
Papaveraceae	0	0	2	2	0	3	7
Campanulaceae	0	1	1	0	0	2	12
Caprifoliaceae	0	0	2	1	0	2	6
Chenopodiaceae	0	0	1	2	0	2	12
Oxalidaceae	2	0	0	1	1	2	3
Plantaginaceae	2	0	2	1	1	2	4
Primulaceae	0	0	2	0	0	2	6
Violaceae	1	0	2	0	0	2	11
Apiaceae	0	0	1	0	0	1	32
Balsaminaceae	0	0	1	0	0	1	3
Boraginaceae	0	0	1	0	0	1	18
Ericaceae	0	1	1	0	0	1	9
Hypericaceae	1	0	1	0	0	1	3
Liliaceae	1	0	0	0	0	1	11
Lythraceae	1	1	1	1	0	1	2
Rhamnaceae	0	1	0	0	0	1	2
Salicaceae	0	0	0	0	1	1	17
Saxifragaceae	0	1	0	1	0	1	7
Tiliaceae	0	0	1	0	0	1	1
Urticaceae	1	1	1	1	1	1	2

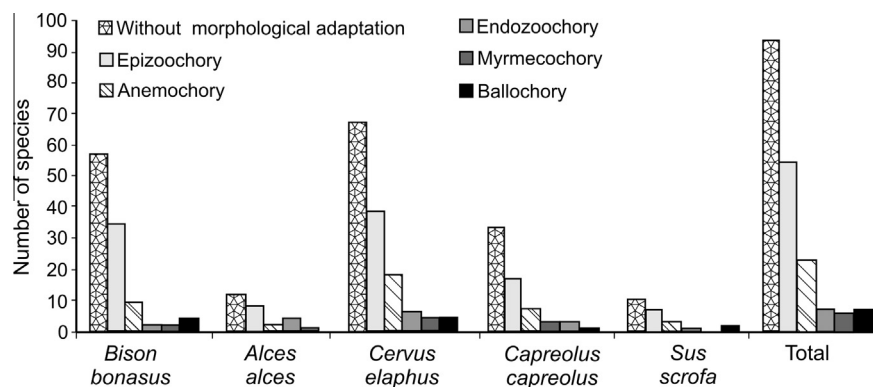


Fig. 1. Number of plant species dispersed endozoochory in Białowieża Forest (north-eastern Poland) by the guild of ungulates: red deer *Cervus elaphus*, European bison *Bison bonasus*, roe deer *Capreolus capreolus*, Eurasian moose *Alces alces* and wild boar *Sus scrofa*, grouped by dispersal adaptations.

that some infrequently dispersed species may not have been detected for bison and red deer, while estimations for moose and wild boar were two times higher than the number of species emerged from their faeces (Table 2).

Only *Juncus effusus* L., *Lycopus europaeus* L., *R. idaeus* and *U. dioica* were present in the faeces of all five vectors. Close to half (46.1%) of all dispersed species were exclusive to the faeces of only a single vector. Over 10% of plants dispersed by bison were those of ancient forest species (Table 2; Appendix B). Bison and red deer dispersed a notably high number of plant species exclusive to them

(29.4% and 29.9% of plant species dispersed, respectively; Table 2). However, most of the exclusive species emerged only from a single dung sample and with low abundance (Appendix A). Regression analysis did not show any dependence between traits of seeds (length, width and mass) and their abundance in the dung of ungulates (all $R^2 < 0.06$).

Plants from 30 families, out of 35 recorded, were dispersed by red deer, while the number of families dispersed by moose and wild boar was two times lower (Table 2). The strongest correlations between the species richness of families dispersed by ungu-

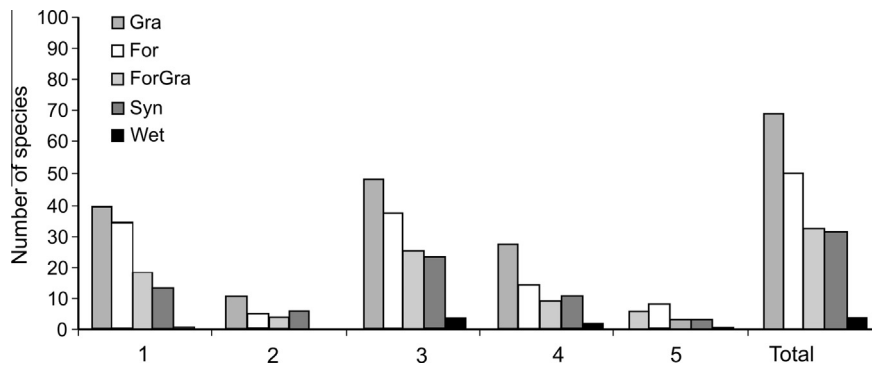


Fig. 2. Number of plant species dispersed endozoochorically in Białowieża Forest (north-eastern Poland) by the guild of ungulates: red deer *Cervus elaphus*, European bison *Bison bonasus*, roe deer *Capreolus capreolus*, Eurasian moose *Alces alces* and wild boar *Sus scrofa*, grouped by habitat of origin: For – forests, Gra – grasslands, Syn – synanthropic (fields, ruderal sites, invasive species), ForGra – species occurring in forest and grassland habitats, Wet – wetlands.

Table 4
Results of ANOSIM analysis of the plant species compositions dispersed by the guild of ungulates in Białowieża Primeval Forest (northeastern Poland); value of *R* statistic above the diagonal, *P* value below the diagonal.

	European bison	Eurasian moose	Red deer	Roe deer	Wild boar
European bison	X	0.3795	0.2422	0.0960	0.1563
<i>Bison bonasus</i>					
Eurasian moose	<0.0001	X	0.6497	0.2353	0.0439
<i>Alces alces</i>					
Red deer	<0.0001	<0.0001	X	0.2356	0.2760
<i>Cervus elaphus</i>					
Roe deer	0.0014	<0.0001	<0.0001	X	0.1069
<i>Capreolus capreolus</i>					
Wild boar	<0.0001	0.0608	<0.0001	<0.0001	X
<i>Sus scrofa</i>					

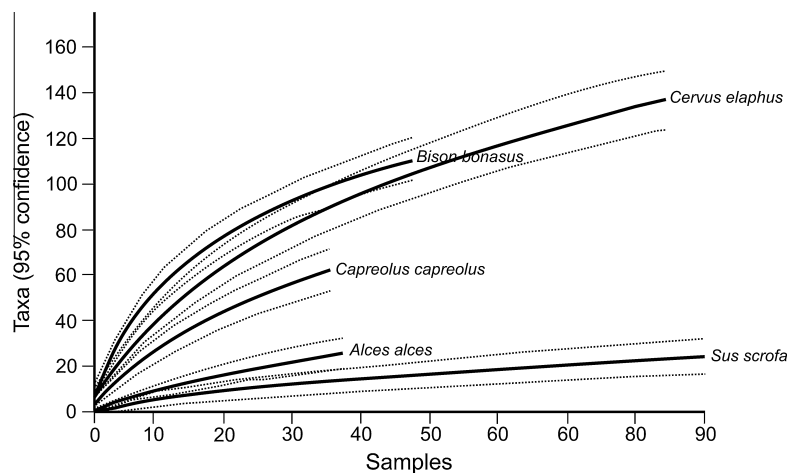


Fig. 3. The cumulative number of plant species dispersed endozoochorically in Białowieża Forest (north-eastern Poland) by the guild of ungulates: red deer *Cervus elaphus*, European bison *Bison bonasus*, roe deer *Capreolus capreolus*, Eurasian moose *Alces alces* and wild boar *Sus scrofa*, with increasing sample size (from sample-based rarefaction). For each species 95% confidence limits are shown; for Chao2 estimators of total species richness see Table 2.

lates and the species richness of these families in Białowieża Forest were observed for roe deer ($r_s = 0.826$; $P < 0.0001$), red deer ($r_s = 0.781$; $P < 0.0001$) and bison ($r_s = 0.739$; $P = 0.0001$), while correlations for wild boar ($r_s = 0.538$; $P = 0.04$) and moose ($r_s = 0.595$; $P = 0.02$) were weaker.

4. Discussion

Our results show that each animal species of the guild of ungulates in close-to-natural European forest plays an important role in

the process of plant dispersal. The set of plant species dispersed by the studied seed vectors was unique to each of them owing to the large proportion of exclusive species and low similarity between the species composition of dispersed plants.

Red deer was the most efficient seed vector among the studied ungulates. Over 95% of its dung samples contained seeds. Red deer dispersed the highest number of plant species (2/3 of all emerged species), out of which close to 30% were species not dispersed by other studied ungulates. These findings are in line with the conclusions of Eycott et al. (2007), who found the red deer/fallow deer complex to be the most efficient seed vector among the studied

herbivore species in Thetford Forest (England). Red deer could potentially replace almost the whole guild of herbivores living in Thetford Forest, dispersing 95% of species emerged from dung samples overall. In Białowieża Forest red deer was also the most efficient disperser, although without the potential to replace the rest of the guild because the proportion of species dispersed by it was much lower than that reported from England. The results of the ANOSIM analysis showed that the sets of species dispersed by the least efficient ungulate dispersers, like moose, are the least similar to any other studied vector, stressing their importance for plant dispersal. Taking these results into account, and also the differences in behaviour and habitat use between the studied seed vectors, resulting in the deposition of seeds in different (micro)habitats (D'hondt et al., 2012), the species diversity of seed dispersers is of paramount importance for plant dispersal.

It needs to be stressed that over half of overall plant species emerged from dung samples were dispersed by European bison – a species exterminated in most European forests centuries ago. Close to 30% of plant species were dispersed exclusively by this ungulate. Thus, we may hypothesise that plant dispersal in European forests has been seriously impoverished since bison extermination. It is worth noting that it dispersed the highest number of ancient forest species (*sensu* Dzwonko and Loster, 2001; Appendix B), and several species of them were dispersed exclusively by this ungulate: *Carex pilosa* Scop., *Circaea lutetiana* L., *Gagea lutea* (L.) Ker Gawl., *Galeobdolon luteum* Huds., *Geum urbanum* L., and *Poa nemoralis* L. (Table 2; Appendix A). The ancient forest species are known to disperse very slowly in present-day forests (Hermy et al., 1999). Is the impaired seed dispersal of these species caused by their biology or by the loss of specific dispersers? This question is very important in the light of reports already showing climate change affecting forest plant biology and expected poleward plant migrations (Sparks et al., 2009).

It is notable that in BPF each of the studied ungulates dispersed a higher number of plant species than reported in any previous papers (compare Eycott et al. (2007) for red deer and roe deer, Heinken et al. (2002) for red deer, roe deer and wild boar). This effect can be attributed to the high plant species and habitat diversity of BPF (Faliński, 2003), and/or to the methodological differences in collecting and handling samples. In most papers dealing with endozoochory dung samples were collected only during the few months of the growing season, dried (or kept in a refrigerator) and ground before mixing with sand/soil, and most of the authors applied the Ter Heerdt method (Ter Heerdt et al., 1996) with short term (up to 6 months) observation of seedling emergence under controlled conditions (e.g. Malo and Suárez, 1995; Cosyns et al., 2005; Eycott et al., 2007). We assume that some elements of this method, e.g. drying, storing, grinding, short period of sample collection and short period of seedling emergence monitoring, can negatively influence the number of registered seedlings and species. We collected dung samples of all ungulates year-round, mixed fresh faeces with sterilized sand, and monitored seedling emergence in an unheated greenhouse for three years (a method similar to the common garden experiment). Wessels and Schwabe (2008) compared the Ter Heerdt method and the common garden method, and reported more species (e.g. monocotyledons) and a higher seedling emergence of hard-seeded (i.e. physically dormant) species of the families Fabaceae and Cistaceae from dung samples in the latter. The common garden method gives seeds more time to come out of dormancy, and fluctuations in abiotic conditions act as a signal to induce water uptake and germinate (Wessels and Schwabe, 2008), which partly explains the higher dispersal characteristics obtained in our study.

Grazers (red deer and bison) dispersed more species than browsers (roe deer and moose), similar to that reported by Mouissie (2004), and each of them dispersed more species than the

omnivorous wild boar. We noted the largest dissimilarities between sets of dispersed plant species for moose compared to all other ungulates. This is probably because of moose preference for wet habitats and foraging on the shoots of trees and bushes, whose share in its winter diet can exceed 98% (Heptner and Nasimovich, 1967), resulting in low seed intake. In effect moose, the second largest member of the studied guild of ungulates, dispersed the lowest number of seeds and species. Thus, in our opinion, the differences between the vectors in the species richness of dispersed plants and seed density in dung reflect not body size, as suggested by Malo and Suárez (1995) but rather the animals' dietary preferences, digestive physiology and habitat preference rather.

Aside from a few plant species preferred by animals and well adapted to endozoochory (many viable seeds recorded in many dung samples of most of the studied ungulates: *U. dioica*, *J. effusus*, *R. idaeus*), single seeds of many plant species could be swallowed serendipitously, as contamination of the main bulk of food, as suggested by Janzen (1984), or as an effect of some rare feeding behaviour of the studied ungulates. The large proportion of plant species emerged from single dung samples and with low seed abundance would support this statement. Random seed intake is also suggested by the observation that the highest number of plant species dispersed by the studied ungulates belonged to the richest plant families in the forest and were correlated with the species diversity of plant families in the forest. The lack of correlation between the frequency of plant species in dung samples and in the field could be counted against this conclusion, but species frequency in the field is not the same as seed accessibility. The species most frequently emerging from dung samples were those yielding high numbers of seeds per individual (e.g. *U. dioica*, *J. effusus*, *Chenopodium album* L., *Lythrum salicaria* L.), which increases the chances of seed intake by animals, and for the survival of some seeds during their passage through animals' digestive tracts.

The high proportion of plant species with seeds without any morphological adaptations to long distance dispersal corroborates the Janzen (1984) 'foliage is the fruit' hypothesis. These species comprised close to half of all species emerged from dung samples during our study, and are commonly counted as species building persistent soil seed banks. Many seeds of these species stay viable in decomposing dung and pass to the forest soil seed banks, increasing their species richness and abundance, and extending the influence of endozoochory for many years after dispersal (Jaroszewicz, 2013). The second most abundant group were species adapted morphologically to epizoochory. It is possible that they developed double zoochoric adaptation: to epi- and to endozoochory, which is in line with the recent emphasis on the importance of polychory, i.e. dispersal by multiple mechanisms (Hampe, 2004; Ozinga et al., 2004; Nathan, 2007; Dennis and Westcott, 2007; Nathan et al., 2008).

In contrast to the dominance of forest habitats at the study site, with only a 5% share of open areas (Sokołowski, 2004), only about one quarter of emerged species were those of forest habitats, and those of diverse grasslands and synanthropic habitats comprised over half of species overall. This probably expresses the preference of large herbivores to feed in clearings (Kuijper et al., 2009), and was already reported from BPF for European bison (Jaroszewicz et al., 2009). It should be noted that most herbivores prefer to forage in open areas but then rest under cover of the forest (Kuijper et al., 2009). Thus, a large proportion of swallowed seeds, including those originated from non-forest habitats, are probably deposited with dung in animal's forest refuges. This effect is already visible in the distribution and abundance of some invasive species in BPF. The domestic apple *Malus domestica* Borkh. s.l. – one of the most common woody invasives in the forest (Adamowski et al., 2002) emerged in our study from the faeces of all the studied ungulates except wild boar. There are many invasive plant species

dispersed endozoochorically (e.g. *Oxalis dillenii* Jacq., *Symphoricarpos albus* Duhamel), or species which do not belong to forest vegetation but may potentially colonize forest roads and disturbed places, e.g. *Trifolium repens* L., *Trifolium dubium* Sibth., *Trifolium hybridum* L., *Stellaria media* (L.) Vill.

5. Conclusions

Our study stresses the importance of the completeness of the guild of seed dispersers for plant dispersal in forest habitats. The high number of plant species dispersed endozoochorically by the studied guild of animals, combined with the low similarity between sets of dispersed plant species, and the high share of species dispersed exclusively by only one of the vectors allows the conclusion that: (1) Dispersal efficiency of ungulates differs, but all species play an important role in the process of plant dispersal in close-to-natural European forest; (2) species diversity of the guild of ungulates is very important for plant dispersal; and (3) plant dispersal in European forests has been seriously impoverished since their guilds of herbivores became deficient.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.foreco.2013.05.004>.

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